

LifeDrop — Blood Donation Network

Complete Technical Project Guide, Newly Added Features & System Architecture

1. Executive Summary & Problem Solved

LifeDrop is a Pakistan-focused voluntary blood donation and hospital emergency coordination network. In Pakistan, acute blood shortages affect over 250,000 thalassemia children, emergency trauma victims, and post-partum mothers annually. The project eliminates middlemen and black marketing by directly matching voluntary donors with patients and hospitals in real time.

Key Technology Stack: Frontend: React 19, TypeScript, Vite, SCSS, React-Router 6, Leaflet Maps, PWA. Backend: Django 6, Django REST Framework (DRF), Daphne ASGI Server, Django Channels (WebSockets), SQLite/PostgreSQL, SimpleJWT Authentication.

2. Newly Added & Upgraded Features (Recent Enhancements)

The following 6 major features were newly designed, developed, and connected to make this project production-grade:

Feature	Module / File	What Was Added & How It Works
1. Local Backend Sync & Dynamic BaseURL	.env api.ts	Fixed inactive remote cloud URL. Created environment-based configuration pointing to active local Django REST API. Enabled media/avatar resolution.
2. Hospital Blood Inventory Center	HospitalPanel.tsx views.py (set_stock)	Interactive reserve management for all 8 blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-). Live +/- unit adjustment buttons, critical low stock alerts (<3 units), and visual progress bars.
3. NGO Blood Camps & Drives Page	CampaignsPage.tsx /campaigns route	Directory of upcoming blood donation camps. City filters (Lahore, Karachi, Islamabad), event timings, perks (Free screening, Refreshments), 1-click volunteer registration, and a 'Host a Drive' modal.
4. WhatsApp Emergency Quick-Contact	SearchBloodPage.tsx DonorProfilePage.tsx	Added dedicated WhatsApp action buttons on verified donor cards and profile pages. Formats instant urgent coordination message with donor name, city, and needed blood group.
5. Digital Certificate of Appreciation	DonationHistory.tsx donations.data.ts	Automated LifeSaver award certificate generation for completed donations. Renders donor name, donated volume, date, hospital, gold seal, and Print/Save PDF support.
6. Realtime WebSockets Chat & Live GPS	Messages.tsx ActiveRequests.tsx	Direct Daphne/Django Channels WebSocket communication in the Messages tab with connection indicators. Added HTML5 GPS location sharing for donors pledging to donate.

3. Total Complete Feature Matrix

A. Donor Discovery & Matching:

- **Smart Compatibility Matching:** Matches both exact blood groups and medically compatible donor groups (e.g. O-universal donor, AB+ universal recipient).
- **Interactive Leaflet Map:** Visualizes nearby donors across Pakistani cities with distance calculation (km).
- **90-Day Medical Cooldown:** Automatically flags donors in cooldown rest periods to prevent unsafe donations.
- **Direct Call & WhatsApp Action:** Instant contact initiation without friction.

B. Emergency Blood Request Workflow:

- **Urgency Priority Levels:** Routine (48-72 hrs), Urgent (12-24 hrs), and Critical (<2 hrs).
- **Auto-Expiration Daemon:** Background checks auto-expire stale requests to keep the feed fresh.
- **Progressive Units Tracking:** Visual bar showing units fulfilled vs units needed.
- **Pledge & Verification Lifecycle:** Donor RSVP -> Live GPS Share -> Hospital/Patient mark as Received.

C. Hospital Command Center & Blood Bank:

- **Stock Reserves Module:** Real-time inventory tracking for all blood groups with critical threshold badges.
- **Predictive Dengue Season Analytics:** AI/ML model alerting hospitals of upcoming regional platelet shortages.
- **Hospital Requests Dashboard:** View and verify blood bags received for admitted patients.

D. User Experience, PWA & Security:

- **QR Code Digital Health Card:** Base64 QR code with donor history, badge, and blood group for instant hospital verification.
- **Progressive Web App (PWA):** Installable on Android & iOS mobile devices with offline service worker caching.
- **Role-Based Authentication:** Dedicated roles for Donors, Patients, and Hospitals secured via JWT tokens.

4. How The Project Works (Step-by-Step Workflow)

Step 1: User & Donor Onboarding

A donor registers on /auth with their email, city, blood group, and phone number. Their profile is created with eligibility checks, cooldown dates, and an initial donor badge.

Step 2: Emergency Request Creation

A patient or hospital creates an emergency request on /request specifying patient name, hospital, city, blood group, units, and urgency. Django signals auto-trigger email alerts and mock social media broadcasts to matching local donors.

Step 3: Discovery & Donor Matching

Users locate donors on /search by searching cities or blood groups. The priority algorithm sorts donors by physical proximity, donation recency, and response rating.

Step 4: Real-time Coordination & Live Tracking

The requester contacts the donor directly via Phone, 1-click WhatsApp, or Daphne WebSocket chat. When the donor clicks 'I will donate', they can activate 'Share GPS Location', sending live coordinates to the hospital/patient.

Step 5: Donation Verification & Gamification

Once the donation is completed at the hospital, the patient or hospital marks it as 'Received'. The donor's total donation count increases, badges upgrade (Bronze -> Silver -> Gold), and an official Digital Certificate of Appreciation is generated.

Step 6: Hospital Blood Bank Management

Hospital administrators manage their facility's blood inventory via the Hospital Command Center, adjusting available units and receiving automated low-stock warnings.

5. Local Execution & Credentials

Frontend: Run `npm run dev` inside LifeDrop/ (Runs at `http://localhost:5173`).

Backend: Run `python manage.py runserver` inside LifeDrop/backend/ (Runs at `http://127.0.0.1:8000/api/`).

Demo Accounts:

- Hospital Admin: `admin@lifestream.com` / Password: `Admin@1234`
- Standard Donors: `salman@gmail.com`, `ahmad@gmail.com`, `amna@gmail.com` / Password: `password123`